

Robot Part Iteration Log

Instructions

ROBOT PART ITERATION LOG

Total: 20 points

Name: Your NAME Group: 6/3 Part Name: MOTOR MOUNT

Part Version # 4 Manufacturing Technique: 3D CNC OTHER (2 pts)

1. WHAT DID YOU CHANGE? (draw or write) (4 pts)

Sketch a quick "before → after" or list 1-2 bullets

- hole size for face mount: 3.3 mm → 3.5 mm
- offset for gearbox: 0.25 mm → 0.30 mm
- WALL Thickness in 3d print: 2 walls → 4 walls

2. CALIPER MEASUREMENTS (pick YOUR 3 critical spots) (6 pts)

Spot (describe)	Target	Actual	± Error	Initials
1 Face mount holes (3mm clearance)	3.3mm	3.35mm	+ 0.05mm	MB
2 Gearbox inside wall to wall	38.75mm	38.65mm	- 0.10mm	MB
3 Face mount MATERIAL THICKNESS	4.00mm	4.03mm	+ 0.03mm	MB

3. FIT / FUNCTION TEST (4 pts)

Identify two features or qualities to evaluate fit and function:

1. Gearbox should fit snug, but plastic must not crack or break
Results: It fits snugly, minimal rotation, no breakage
2. Mounting holes (to deck): Screws should slip-fit and heads don't interfere
Results: slip fit, no interference

4. OBSERVATION + NEXT MOVE (4 pts)

One sentence: what surprised you:

Very snug fit and it still didn't break. Prob. because of increased # of walls in 3d print.

One sentence: what you'll change for v₊₁:

I'd like to change orientation of the print and evaluate overall strength.

"Remember kids, the only difference between screwing around and science is writing it down."

- Adam Savage, Myth Busters

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- Fill out the header
- Indicate the version number of your part
- Indicate your manufacturing technique

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Part Version # 4 Manufacturing Technique: 3DP CNC OTHER (2 pts)

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- This part is filled out while you are manufacturing your part (before evaluation).
- If you only changed one feature, you must be very detailed about why (i.e., previous version have proven success of all other features)

1. WHAT DID YOU CHANGE? *(draw or write)* (4 pts)

Sketch a quick "before → after" or list 1-2 bullets

- hole size for face mount : 3.3 mm → 3.5 mm
- offset for gearbox : 0.25 mm → 0.30 mm
- WALL Thickness in 3d print : 2 WALLS → 4 WALLS

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- You should know ahead of time what features you're evaluating.
- When your part is completed, measure three features to evaluate tolerance
- Error equals Actual Measurement minus Target Value

2. CALIPER MEASUREMENTS (pick YOUR 3 critical spots) (6 pts)

Spot (describe)	Target	Actual	$\pm$ Error	Initials
1 Face mount holes (3mm clearance)	3.3mm	3.35mm	+ 0.05mm	MB
2 Gearbox inside wall to wall	38.75mm	38.65mm	- 0.10mm	MB
3 Face mount material thickness	4.00mm	4.03mm	+ 0.03mm	MB

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- How does it ACTUALLY fit?
- Record the metric you will use to evaluate fit/function, and then record the results.

3. FIT / FUNCTION TEST (4 pts)

Identify two features or qualities to evaluate fit and function:

1. Gearbox should fit snug, but plastic must not crack or break Results: It fits snugly, minimal rotation, no breakage
2. Mounting holes (to deck): Screws should slip-fit and heads don't interfere Results: Slip fit, no interference

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- Where will you go from here? Is the part ready for service or do you need to test another version?

4. OBSERVATION + NEXT MOVE (4 pts)

One sentence: what surprised you:

Very snug fit and it still didn't break. Prob. because of increased # of walls in 3d print.

One sentence: what you'll change for v_+1:

I'd like to change orientation of the print and evaluate overall strength.